Dating Apps Reviews

Alberti Luca (1236389)
Canton Matteo (1236962)
Strepparava Miriam (1232275)

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Abstract

This report presents a computational text analysis of user reviews from three major dating applications: Tinder, Bumble, and Hinge. The study focuses on reviews posted in 2021 and aims to compare user perceptions across platforms by combining sentiment analysis and topic modeling. Based on a publicly available Kaggle dataset, an initial balanced sample of 15,000 reviews was extracted, with 5,000 reviews for each application. The reviews containing fewer than 7 words were then removed before the document-term matrices used for topic modeling. After preprocessing the textual data, sentiment analysis was performed using a lexicon-based approach based on the Bing dictionary, while topic detection was carried out through the Mixture of Unigrams model. The optimal number of topics was selected separately for each application using the AIC criterion, resulting in 12 topics for Hinge, 14 for Bumble, and 13 for Tinder. The topics were then aggregated into broader macro-topics to improve interpretability and enable comparison across platforms. The results show that negative sentiment is prevalent across all three applications, although the main drivers of dissatisfaction differ. Common issues include payments, fake profiles, account bans, technical problems, and app functionalities. Hinge is mainly affected by fake users or bans and technical issues, Bumble by fake profiles and money-related dissatisfaction, and Tinder by account bans, payments, and scam profiles.

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1 Introduction & Objective

The online dating industry experienced growth and a shift in user dynamics during the early 2020s, largely influenced by global pandemic restrictions. Among the leading platforms, Tinder, Bumble, and Hinge emerged as the primary channels for digital romance. Understanding user satisfaction and behavior during this critical period provides valuable insights into consumer psychology and platform performance. This report focuses on the analysis of user-generated data throughout the year 2021, a pivotal time when digital interactions heavily replaced traditional socializing.
The primary objective of this study is to perform a comparative analysis of Tinder, Bumble, and Hinge to uncover how users perceived and interacted with each app during 2021. To achieve this, our methodology is split into two main analytical phases:
- Sentiment Analysis: We evaluate the emotional tone of user reviews across all three platforms to measure overall satisfaction levels, tracking positive, negative, and neutral sentiments.
- Topic Detection: To understand why users felt a certain way, we perform topic modeling using the Mix of Unigrams (MoU) model. This allows us to extract the main themes, recurring complaints, and unique features praised by each application.

2 Dataset Description

The analysis is conducted using an open-source dataset publicly available on Kaggle, namely the "Dating App Reviews" dataset: `https://www.kaggle.com/datasets/sidharthkriplani/datingappreviews/data`(last accessed on May 25, 2026). In its original form, the dataset comprised 681,986 observations and 8 variables.

The variables included in the raw data set are structured as follows.

- `X`: The row index or unique identifier for each observation in the dataset.
- `Name`: The username or profile name of the reviewer.
- `Review`: The textual content of the review left by the user.
- `Rating`: The score given to the app, typically on a scale from 1 to 5 stars.
- `X.ThumbsUp`: The number of times other users marked the review as helpful or upvoted it.
- `Date.Time`: The specific date and time the review was published.
- `App`: The specific dating platform the review refers to (Tinder, Bumble, or Hinge).
- `Year`: The calendar year in which the review was posted, extracted from the `Date.Time` variable.

To perform a targeted and computationally efficient analysis, a subset of data was extracted from the main data set using a specific sampling strategy in R applied uniformly to all three platforms (Tinder, Bumble, and Hinge). The process was structured as follows.

- **Temporal and Platform Filtering:** The dataset was initially filtered to isolate observations from the calendar year 2021 (`Year == 2021`). From this subset, data was partitioned into three distinct subsets based on the platform (`App == "Tinder"`, `App == "Bumble"` and `App == "Hinge"`).
- **Random Sampling:** A reproducible simple random sampling without replacement was performed on each platform's data using a fixed random seed (`set.seed(1234)`). An initial balanced sample of 5,000 reviews was extracted for each of the three applications.

- **Text Length Filtering:** After sampling, reviews containing fewer than 7 words were excluded. This step removes very short and non-informative reviews, such as "Good" or "Great app", which do not provide enough textual content for meaningful topic detection. As a result, the datasets used for the construction of the document-term matrices contain slightly fewer than 5,000 reviews for each application.

The resulting samples are characterized by the parameters summarized in Table 1.

Table 1: Characteristics of the Sample

Characteristic	**Description**
Platforms Covered	Tinder, Bumble, and Hinge
Temporal Window	The entire calendar year of 2021
Initial Sample Size per App	5,000 reviews before text-length filtering
Minimum Text Length	At least 7 words per review for topic modeling
Final Textual Sample	Slightly fewer than 5,000 reviews per app after filtering
Variables Extracted	Textual content (`Review`) and engagement metric (`X.ThumbsUp`)

3 Data Preprocessing

To prepare textual data for topic modeling and sentiment analysis, a rigorous text preprocessing function was implemented. The procedure consists of the following sequential steps:

- **Corpus Creation:** The raw text vectors are converted into a Volatile Corpus (`VCorpus`) using a `VectorSource`. This structured environment allows efficient text manipulation within the `tm` framework.

- **Case Normalization:** All text is converted to lowercase (`tolower`). This ensures that the algorithm treats identical words uniformly, regardless of their position in a sentence (e.g., treating "App" and "app" as the same token).

- **Punctuation and Number Removal:** Punctuation marks (`removePunctuation`) and numerical digits (`removeNumbers`) are stripped from the text. These elements generally do not contribute to the semantic meaning of user reviews and are treated as noise.

- **Lemmatization:** Words are reduced to their base or dictionary form (e.g., "running", "runs", and "ran" are all mapped to "run") using the `lemmatize_strings` function. This step groups inflected forms of a word so that they can be analyzed as a single item, drastically improving topic clustering performance.

- **Stopword Removal:** Standard English stopwords provided by the `tm` package (`stopwords("english")`) are removed. This cleaning ensures that any remaining functional words (such as pronouns, prepositions, and conjunctions) or newly uncovered base forms are completely purged.

- **Whitespace Stripping:** Extra spaces, tabs, and blanks generated by the previous removal steps are eliminated (`stripWhitespace`), collapsing multiple spaces into a single space.

- **Term-Document Matrix (TDM) Generation:** The cleaned corpus is transformed into a Term-Document Matrix (`TermDocumentMatrix`), which maps the frequency of terms across all sampled reviews.

- **Sparsity Reduction:** To improve computational efficiency and reduce the curse of dimensionality, extremely rare terms are filtered using `removeSparseTerms` with a threshold of 0.995. This keeps only the terms that appear in at least 0.5% of the documents, effectively removing unique typos or highly idiosyncratic words while retaining the most meaningful vocabulary. The following table shows the dimensions of each term-document matrix

Table 2: Dimensions of the Term-Document Matrices after text-length filtering (p: Terms, n: Documents) for each analyzed dating application.

Application	**Terms** (p)	**Documents** (n)	**Dimensions** (p, n)
Hinge	432	4794	$(432, 4\,794)$
Bumble	391	4725	$(391, 4\,725)$
Tinder	358	4601	$(358, 4\,601)$

4 Exploratory Text Analysis

Following preprocessing, the data are structured to enable exploratory visual analysis. A data frame is constructed to pair each unique term with its calculated absolute frequency. The preprocessing function concludes by returning a structured list containing the processed matrix, the word frequencies, the cleaned corpus, and the raw text sample, along with an embedded custom function designed to generate a word cloud visualization. This visualization relies on the `wordcloud` package and is heavily customized through specific parameters to ensure clarity and interpretability. To eliminate background noise, the function suppresses rare words by requiring a minimum frequency of 5 appearances. Furthermore, it avoids visual clutter by strictly capping the display at the top 100 most frequent terms. Instead of placing these terms at random, the algorithm enforces a deterministic hierarchical layout, embedding the most prominent words right at the center of the cloud while radiating less frequent terms toward the margins. The font sizes are carefully balanced using a scale ranging from 3 for the maximum frequency down to 0.5 for the minimum, which establishes an immediate graphical hierarchy. Finally, the visualization applies a high-contrast qualitative palette from the `RColorBrewer` library, specifically using eight colors from the “Dark2” set, which enhances readability and allows the distinct thematic clusters of words to stand out effectively.

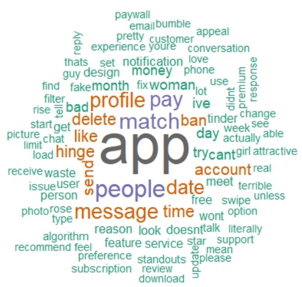

Figure 1: Word Cloud for Hinge.

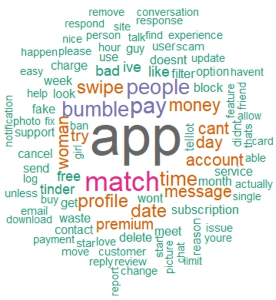

Figure 2: Word Cloud for Bumble

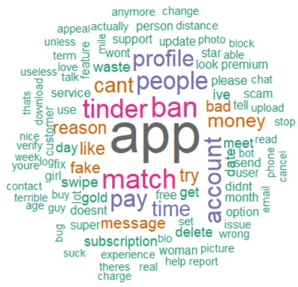

Figure 3: Word Cloud for Tinder

An examination of Figure 1 reveals that Hinge's most frequent terms are *app*, *people*, *match*, *pay*, and *profile*, reflecting both the relational purpose of the app and recurring concerns around monetization. The high frequency of *message* and *date* is consistent with Hinge's focus on meaningful connections. However, *ban*, *delete*, and *money* signal persistent frustration with account moderation and subscription costs.

Figure 2 presents the word cloud for Bumble. The most frequent terms are *app*, *match*, *pay*, *people*, and *time*, reflecting the core user experience of the platform. Notably, *swipe*, *woman*, and *premium* rank among the top words, pointing to Bumble's female-first mechanic and its monetization structure. The presence of *money*, *bad*, and *cant* further indicates a recurring layer of user frustration.

Lastly, Figure 3 shows Tinder's word cloud. The dominant terms are *app*, *ban*, *match*, *people*, and *pay*, where *ban* stands out as the second most frequent word overall a pattern not observed in Hinge or Bumble. Alongside these, *money*, *fake* and *bad* form a strong cluster of negative terms, indicating widespread dissatisfaction with account management, profile authenticity, and subscription costs.

5 Sentiment Analysis

Sentiment analysis represents an algorithmic approach to identifying and processing human emotions, perspectives, and tones embedded within textual data. The primary objective of this technique is to systematically evaluate the text and categorize its underlying disposition as positive, negative, or indifferent. Within the field of text analytics, this methodology plays a pivotal role and is widely deployed to decode user reviews, evaluate consumer feedback, track brand reputation on social media networks, and analyze shifts

in general public perception. The main approaches are the following.
Supervised Machine Learning: Utilizes labeled datasets to train statistical classifiers (such as SVM, logistic regression, or neural networks).
Lexicon - based Methods : match words to sentiment dictionaries (e.g., positive or negative words)
In this project, the lexicon-based method approach was used. To perform this analysis, the Bing lexicon was utilized, which is a pre-compiled dictionary that maps thousands of English words to a dichotomous emotional polarity (positive or negative).

In the initial stage, tokenization is executed, a process in which each review is decomposed into individual words, or tokens. Subsequently, the data set is filtered using a standard list of stop words, which isolates and retains only the words that carry significant semantic meaning.

Through an `inner_join` operation, the remaining words in the reviews are crossed with the Bing dictionary. This step isolates words that possess an explicit emotional connotation; conversely, neutral words are assigned a value of zero.

Ultimately, this workflow produces a structured table consisting of four columns: `line`, `negative`, `positive`, and `sentiment_score`. The `sentiment_score` is computed mathematically as the difference between the positive and negative scores ($Score = Positive - Negative$). Consequently, a definitive individual sentiment score is calculated for each specific `line` within the sample.
The following three graphs illustrate the distribution of the sentiment scores relative to the total number of reviews (complaints) for each individual application. In addition, reviews that do not contain any initial emotional connotation are reintegrated into the analysis. If a review lacks a sentiment score because none of its words matches the Bing lexicon, its score is deliberately forced to 0, thus classifying it as neutral.

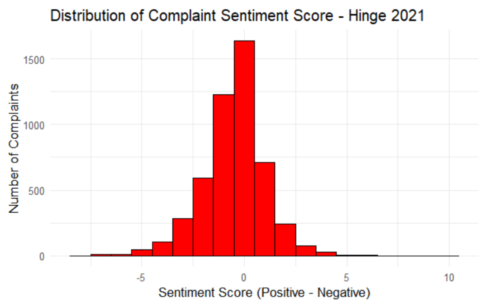

Figure 4: Histogram of Sentiment score for Hinge

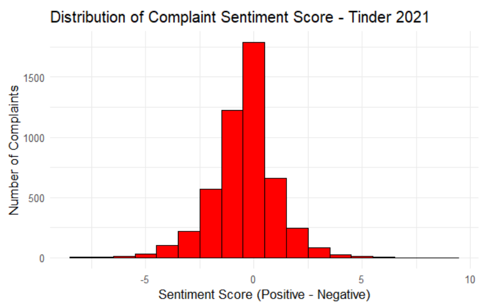

Figure 5: Histogram of sentiment score for Tinder

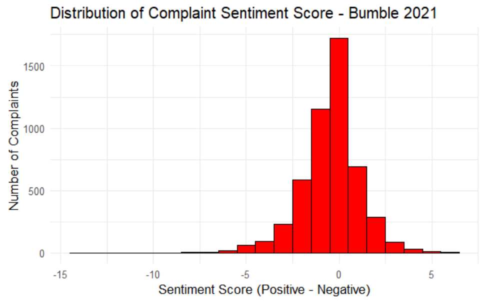

Figure 6: Histogram of sentiment score for Bumble

An examination of the plots reveals that the distribution of sentiment scores follows a highly similar pattern across all three applications. Specifically, the vast majority of user reviews cluster tightly around a zero score. A closer look at the distributions indicates that the left tail is consistently heavier and more extended than the right tail in all three histograms. This subtle asymmetry suggests a slight prevalence of moderately negative reviews compared to highly positive reviews on all platforms.
Subsequently, three bar plots were generated to visualize the categorical sentiment. The x-axis represents the three sentiment classifications (`Negative`, `Neutral`, and `Positive`), while the y-axis displays the total count of complaints for each category. The categorical sentiment variable is derived directly from the computed sentiment score based on strict conditional logic. Specifically, a review is classified as `Positive` if its sentiment score is strictly greater than zero ($Score > 0$), and as `Negative` if the score is strictly less than zero ($Score < 0$). Reviews that initially lacked any emotional score—either because they contained no polarized words within the Bing lexicon or because their positive and negative word counts canceled each other out—are mapped into the `Neutral` category, with missing values explicitly handled and imputed to zero.

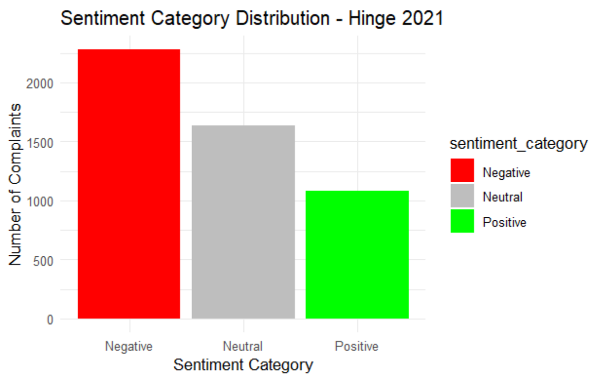

Figure 7: Barplot of Sentiment Categorical Distribution for Hinge

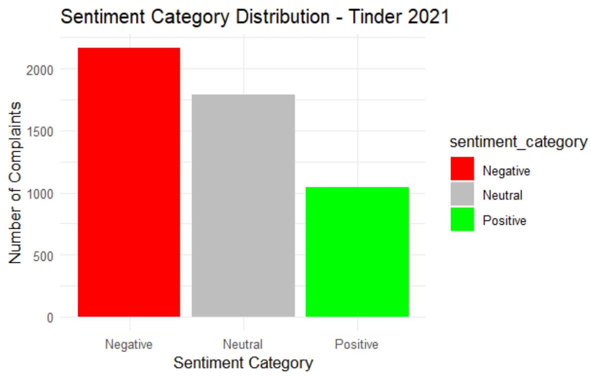

Figure 8: Barplot of Sentiment Categorical Distribution for Tinder

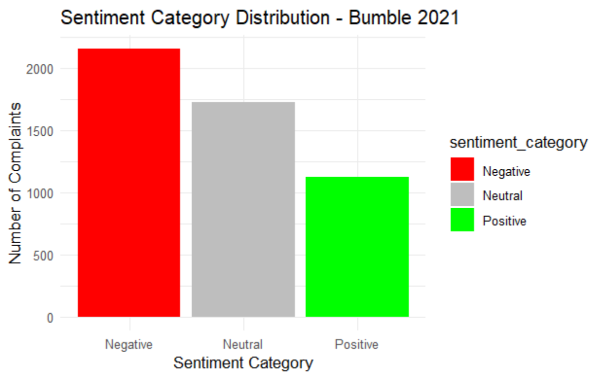

Figure 9: Barplot of Sentiment Categorical Distribution for Bumble

The results indicate that for all three dating apps, Negative feedback represents the dominant category, while positive sentiment is the least dominant.
So, the initial analysis indicates that there is no substantial difference in user sentiment across the three applications.

6 Model for Topics Detection

The purpose of topic detection is to identify the main themes emerging from user reviews and to understand which aspects of each dating application are most frequently discussed. Although the sentiment analysis presented in the previous section measures whether reviews are positive, negative, or neutral, topic analysis allows us to investigate the reasons behind those opinions by grouping reviews according to their textual content.

For this purpose, we applied the *Mixture of Unigrams* (MoU) model separately to the 2021 review samples of Hinge, Bumble, and Tinder. The MoU model is particularly suitable for this task, because it assumes that each document belongs to one main latent topic. This assumption is reasonable in our context, since most app reviews are relatively short and usually focus on a single dominant issue, such as payments, account bans, fake profiles, matching problems, or technical bugs.

6.1 Mixture of Unigrams model

Let X be the document-term matrix, where each row represents a review and each column represents a term. In the Mixture of Unigrams framework, each review x_d is assumed to be generated from a latent topic $z_d \in \{1, \ldots, k\}$. The probability of observing a document is therefore:

$$p(x_d) = \sum_{i=1}^{k} \pi_i p(x_d \mid z_d = i),$$

where π_i is the prior probability of the topic i, and $p(x_d \mid z_d = i)$ follows a multinomial distribution with topic-specific word probabilities. In this way, reviews assigned to the same topic are expected to have similar term distributions.

Before estimating the model, additional preprocessing was applied to create a document-term matrix suitable for topic detection. Reviews containing fewer than 7 words had already been removed before this step. Generic and low-informative words were removed, sparse terms were filtered out, and the final document-term matrix was built using the remaining reviews. After this preprocessing phase, some reviews contained fewer than two informative terms belonging to the retained vocabulary. These reviews were not directly assigned to a latent topic by the Mixture of Unigrams model and were instead included in the final results under the category *Not Classified*. Therefore, the macro-topic summaries refer to the filtered textual sample used for topic modeling, including both classified reviews and reviews assigned to the *Not Classified* category.

The number of topics k was selected by estimating the MoU models for values of k between 2 and 15 and choosing the model with the lowest AIC, a criterion that balances the fit and complexity of the model. This procedure was applied independently to the three applications.

6.2 Selection of the number of topics

Figures 10, 11, and 12 show the AIC values obtained for different numbers of topics. The selected number of topics is the value of k that corresponds to the minimum AIC.

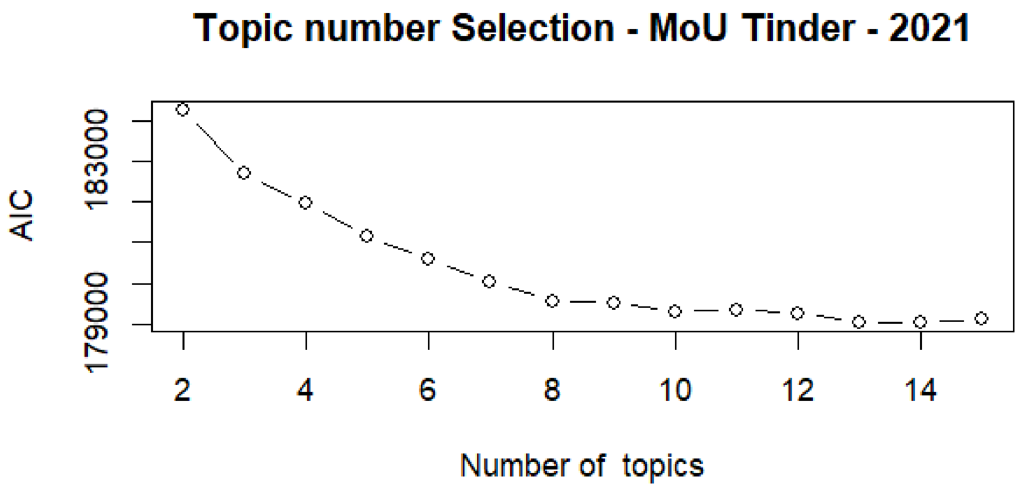

Figure 10: Selection of the number of topics for Tinder using the MoU model.

For Tinder, the AIC decreases sharply for the first values of k and then becomes more stable. The minimum is reached when the model uses 13 topics. These topics were subsequently inspected through their most representative words and then aggregated into broader macro-topics.

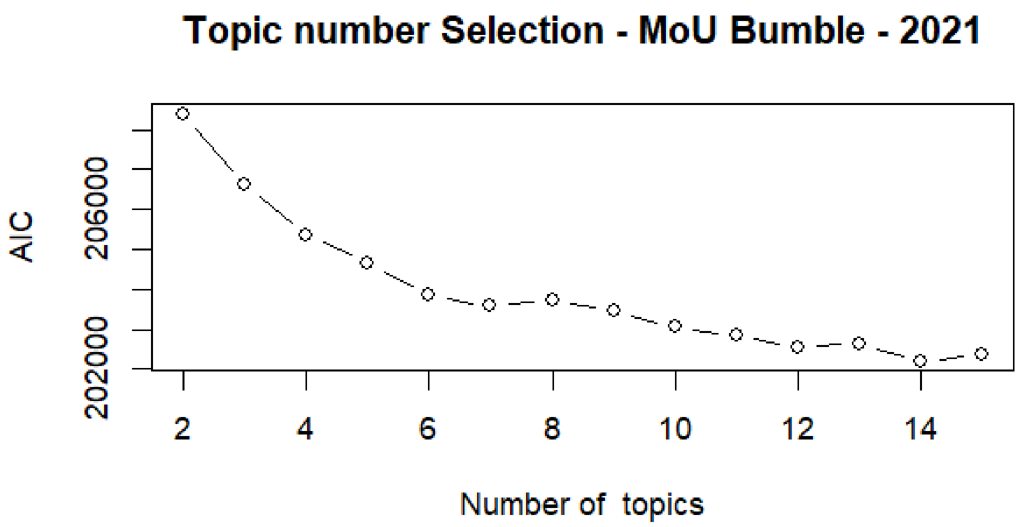

Figure 11: Selection of the number of topics for Bumble using the MoU model.

For Bumble, the decreasing pattern is also visible, although the curve remains less flat than in the other applications. The selected solution contains 14 topics, which were then manually grouped into interpretable macro-topics.

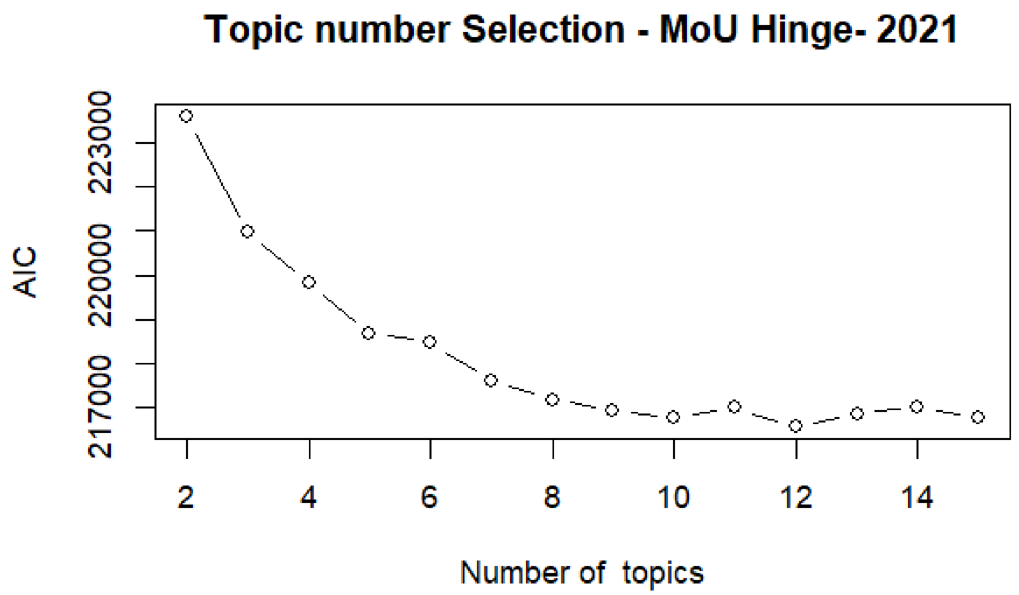

Figure 12: Selection of the number of topics for Hinge using the MoU model.

For Hinge, the AIC reaches its minimum at 12 topics. After estimating the final model, the most frequent and discriminating words within each topic were used to assign a semantic interpretation to the topics.

6.3 From topics to macro-topics

The raw topics obtained from the MoU model are not always directly interpretable as final themes. For this reason, after examining the most representative words and reviews for each topic, similar topics were manually aggregated into broader macro-topics. This step improves interpretability and allows for a clearer comparison between the three dating apps.

The main macro-topics identified across the apps are related to:

- payment systems, subscriptions, premium features, and perceived paywalls;
- account bans, blocked profiles, and verification issues;
- fake profiles, scam accounts, and low-quality users;
- match quality, messaging, ghosting, and conversations;
- swiping, filters, search preferences, and app functionalities;
- technical problems, bugs, login issues, and support.

Although these themes appear in all three platforms, their relative importance differs substantially between apps.

6.4 Hinge results

For Hinge, the largest macro-topic is *Dating*, with 1,031 reviews, followed by *Fake user/Ban* with 894 reviews and *Payments* with 825 reviews. This suggests that Hinge users mainly discuss the quality of the dating experience, the presence of fake users or account bans, and the economic dimension of the app. Other relevant themes are *Not Classified*, with 536 reviews, *Functionalities*, with 524 reviews, and *Technical Problems*, with 509 reviews.

Table 3: Hinge: macro-topic distribution and sentiment counts.

Macro-topic	**Total**	**Negative**	**Neutral**	**Positive**
Dating	1031	440	322	269
Fake user/Ban	894	498	247	149
Functionalities	524	268	172	84
Match/Messages	195	87	57	51
Money Waste	280	168	72	40
Not Classified	536	170	266	100
Payments	825	307	236	282
Technical Problems	509	278	174	57

Figure 13 shows the sentiment composition of each *Hinge* macro-topic. The results reveal that negative sentiment is particularly strong for *Fake user/Ban*, *Dating*, *Technical Problems*, and *Functionalities*. The *Fake user/Ban* macro-topic is especially critical: it is one of the largest themes in terms of volume and contains a much higher number of negative reviews than positive ones. This suggests substantial user dissatisfaction related to fake profiles, account suspensions, bans, and trust or moderation issues.

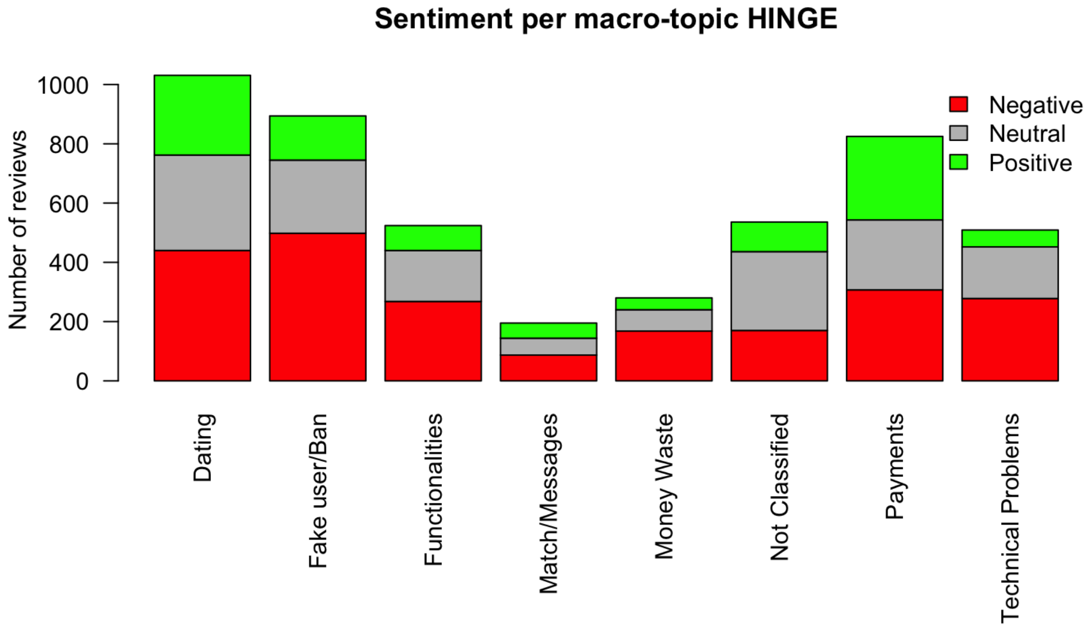

Figure 13: Sentiment by macro-topic for Hinge.

The *Dating* macro-topic is the most frequent overall and also shows a clear prevalence of negative reviews, indicating that many users express dissatisfaction with the dating experience itself, including the quality of matches, user preferences, and the perceived effectiveness of the platform. Similarly, *Technical Problems* and *Functionalities* show a strong imbalance toward negative sentiment, suggesting frustration with bugs, app performance, notifications, messages, filters, and core platform features.

The *Payments* topic is also highly relevant, confirming that subscription costs, premium services, and paywall-related features are central elements in user discussions. However, compared to other macro-topics, it has a more balanced sentiment distribution and also contains a relatively high number of positive reviews. This may suggest that, although pricing and paid features are frequently criticized, some users still perceive premium services as useful or valuable. Finally, the *Not Classified* group is mainly neutral, which is consistent with its more heterogeneous and less clearly defined semantic content.

6.5 Bumble results

For *Bumble*, the most frequent macro-topic is *Payments*, accumulating a total of 911 reviews. Other prominent themes emerged from the analysis, including *Fake/Ban* with 615 reviews, *Functionalities* with 606 reviews, *Match/Messages* with 574 reviews, and *Dating* with 565 reviews.

Table 4: Bumble: macro-topic distribution and sentiment counts.

Macro-topic	Total	Negative	Neutral	Positive
Dating	565	167	170	228
Fake/Ban	615	353	185	77
Match/Messages	574	265	178	131
Not Classified	628	188	302	138
Payments	911	390	313	208
Technical Problems	227	98	99	30
Money Waste	376	232	89	55
Support	223	107	54	62
Functionalities	606	287	204	115

Figure 14 shows that Bumble has a distinctive pattern. The *Dating* macro-topic is the only major topic where positive reviews exceed negative reviews. This indicates that,when users discuss the dating experience itself, Bumble receives comparatively favorable feedback.

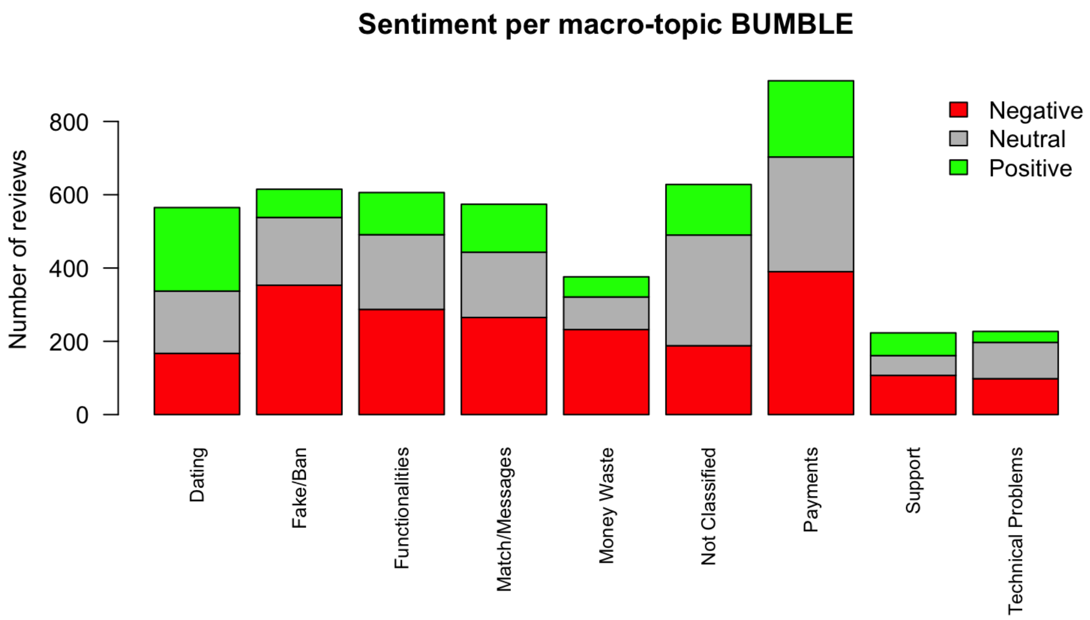

Figure 14: Sentiment by macro-topic for Bumble.

However, the macro-topics *Fake/Ban*, *Payments*, *Functionalities*, and *Money Waste* show a clear prevalence of **negative sentiment**. The strongest imbalance appears in *Fake/Ban*, where negative reviews are significantly more frequent than positive ones. This structural asymmetry strongly suggests that moderation policies, automated account blocking, the proliferation of fake profiles, and general trust-related issues represent a major source of dissatisfaction for *Bumble* users.

6.6 Tinder results

Tinder presents a slightly different structure compared to *Hinge* and *Bumble*. The largest macro-topic is *Ban/Blocked Account*, with 987 reviews, followed by *Payments/Premium* with 866 reviews and *Not Classified* with 657 reviews. This result strongly suggests that account restrictions and premium monetization are particularly central in *Tinder* reviews.

Table 5: Tinder: macro-topic distribution and sentiment counts.

Macro-topic	**Total**	**Negative**	**Neutral**	**Positive**
Ban/Blocked Account	987	479	368	140
Dating	345	110	110	125
Fake/Scam	325	233	58	34
Filters/Search	274	128	105	41
Login/Account	182	83	81	18
Match/Messages	270	144	81	45
Not Classified	657	218	309	130
Payments/Premium	866	311	248	307
Profile/Pics/Bug	311	152	112	47
Money Waste	312	198	77	37
Support	72	28	21	23

Figure 15 confirms that the most problematic area for Tinder is *Ban/Blocked Account.*This topic has the highest number of reviews and the highest number of negative reviews. Users frequently associate Tinder with blocked accounts, bans, login difficulties, and unclear account restrictions.

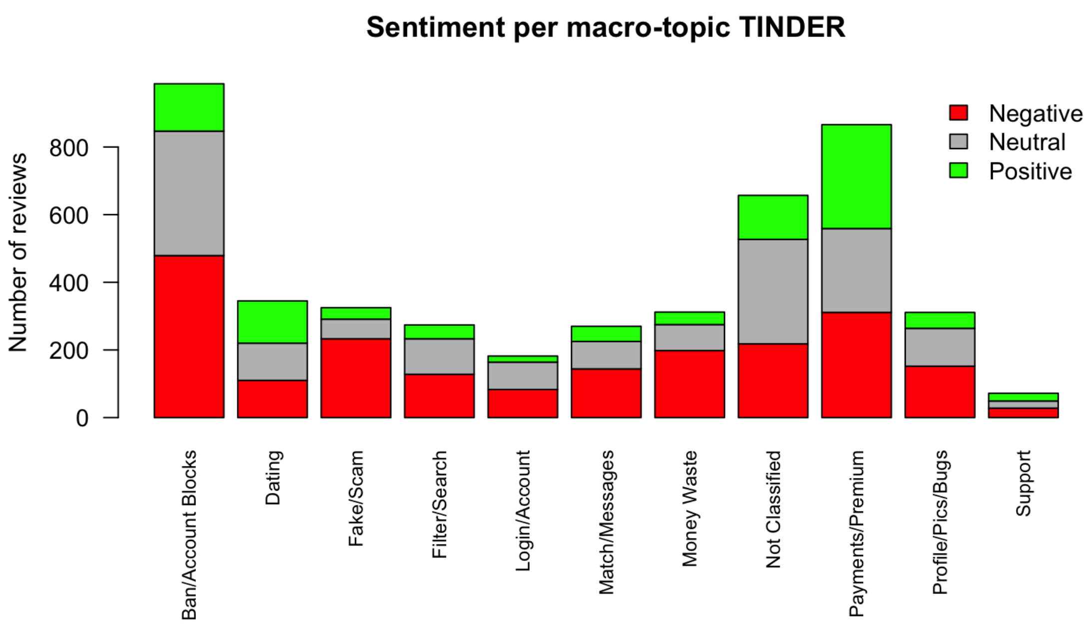

Figure 15: Sentiment by macro-topic for Tinder.

The topic *Payments/Premium* is also highly relevant. Interestingly, it contains almost the same number of negative and positive reviews, indicating a more polarized perception of Tinder's paid services. Some users complain about costs, premium limitations, or perceived unfairness, while others may evaluate premium features more positively.

The *Fake/Scam* and *Money Waste* categories are strongly negative, showing that trust and perceived value are key weaknesses in Tinder's user experience. By contrast, the *Dating* category is relatively balanced and even shows slightly more positive than negative reviews, suggesting that dissatisfaction is not necessarily related to the dating concept itself, but rather to platform management, monetization, account issues, and fake profiles.

6.7 Comparison across applications

The comparison across the three applications highlights several common patterns:

– **First**, payment-related topics are among the most frequent topics for all platforms. This confirms that monetization strategies, subscriptions, premium features, and paywalls are central elements in user evaluations of dating apps.
– **Second**, negative sentiment dominates most operational and technical topics. Complaints are particularly visible in categories related to bans, fake accounts, technical issues, and money waste. These topics are directly related to the trust of users in the platform and the perception of fairness in the service.
– **Third**, the more general Dating topic is not always negative. In Bumble and Tinder, this topic shows a relatively high number of positive reviews. This suggests that users may still appreciate the idea and potential usefulness of dating apps, even when they criticize specific platform mechanisms.

Finally, each app has a distinctive weakness: Hinge is mainly characterized by the *Dating*, *Fake user/Ban*, and *Payments* macro-topics, with negative sentiment especially concentrated in fake users or bans, technical problems, and functionalities. Bumble shows a relevant concentration of negative reviews in fake profiles, bans, and perceived waste of money, but receives comparatively positive feedback on the dating experience. Tinder is especially affected by account bans and blocked accounts, followed by premium/payment issues, and fake or scam profiles. In general, the topic analysis complements the previous sentiment analysis by showing that the negative tone found in many reviews is not generic. Instead, it is concentrated on specific and recurring problems: monetization, account management, fake users, app functionalities, and the perceived effectiveness of matching systems. These aspects represent the main drivers of dissatisfaction in the 2021 reviews of Hinge, Bumble, and Tinder.

7 Conclusions

This project analyzed user reviews of three major dating applications, Tinder, Bumble and Hinge, during 2021. The objective was to compare user perceptions across platforms by combining sentiment analysis and topic modeling. The analysis started from a balanced sample of 15,000 reviews, with 5,000 reviews for each application. After removing reviews with fewer than 7 words, the topic modeling phase was conducted on the remaining textual samples, while reviews with fewer than two informative terms in the final document-term matrix were assigned to the *Not Classified* category.

The exploratory text analysis showed that the three platforms share several recurring terms, such as *app*, *profile*, *match*, *people*, *pay*, and *money*. These words already suggest that user discussions are mainly concentrated around the general app experience, interaction with other users, account management, and monetization. At the same time, platform-specific terms revealed different sources of dissatisfaction: Hinge reviews frequently mentioned dating experience, fake users or account bans, payment-related issues, technical problems, and app functionalities; Bumble reviews showed attention to swiping, blocking, fake profiles, and filters; while Tinder reviews were strongly associated with payments, fake profiles, account bans, and matching problems.

The sentiment analysis provided a first general overview of user satisfaction. Across all three apps, the sentiment score distributions were highly concentrated around zero, meaning that many reviews were classified as neutral or only weakly polarized. However, the left tail of the distributions was consistently heavier than the right tail, indicating a greater presence of negative sentiment. When reviews were classified into negative, neutral and positive categories, negative feedback emerged as the dominant category for all three applications, while positive sentiment was generally less frequent. Therefore, from a purely emotional perspective, the three platforms show a similar overall pattern, characterized by a prevalence of dissatisfaction.

The topic modeling phase allowed us to better understand the reasons behind this negative trend. Using the Mixture of Unigrams model, the optimal number of topics was selected through the AIC criterion separately for each platform. The final models selected 12 topics for Hinge, 14 topics for Bumble, and 13 topics for Tinder. Since the raw topics were sometimes too specific or overlapping, they were manually aggregated into broader macro-topics in order to improve interpretability and enable comparison across apps.

The topic analysis revealed that several issues are common to all three platforms. Payment-related topics are among the most common for every app, confirming the central role of subscriptions, premium features, paywalls, and perceived value for money in user evaluations. In addition, negative sentiment is particularly concentrated on operational and trust-related topics, such as account bans, fake profiles, scam users, technical problems, login issues, and customer support. These results suggest that user dissatisfaction is related not only to the difficulty of finding successful matches but also to the way platforms manage access, monetization, moderation, and reliability.

Despite these common patterns, each app presents a specific profile:

Hinge is primarily characterized by the *Dating*, *Fake User/Ban*, and *Payments* macro-topics. Negative sentiment is most concentrated, in proportional terms, in *Fake User/Ban*, *Technical Problems*, and *Functionalities*, pointing to user dissatisfaction with moderation, app reliability, and core features. However, in absolute terms, the highest volume of negative reviews falls under *Dating*, *Fake User/Ban*, and *Payments*, which are also the largest topics overall.

Bumble shows a strong concentration of negative sentiment around fake profiles, bans, and perceived waste of money. However, its general *Dating* topic receives comparatively more positive feedback, suggesting a more balanced perception of the dating experience itself.

Tinder is especially affected by account bans and blocked accounts, followed by premium payment issues and fake or scam profiles. This suggests that dissatisfaction is mainly related to moderation, monetization, and trust-related problems.

This indicates that while the general sentiment is similar across platforms, the concrete drivers of dissatisfaction differ from one app to another. For Hinge, the main critical areas are fake users or bans, technical problems, and functionalities; for Bumble, fake profiles, bans, and money-related dissatisfaction are particularly relevant; while for Tinder, account bans, payments, and fake or scam profiles represent the most problematic dimensions.

An important result is that the *Dating* macro-topic is not always the most negative one. In Hinge, *Dating* is the largest macro-topic, but the strongest negative imbalance is observed in *Fake user/Ban*, *Technical Problems*, and *Functionalities.* In Bumble and Tinder, the

Dating category appears relatively more balanced and, in some cases, contains a high number of positive reviews. This suggests that users do not necessarily reject the idea of dating apps themselves. Instead, many criticisms are directed toward platform-specific mechanisms, such as unclear bans, aggressive monetization, low match quality, fake users, or poor technical performance. In other words, the negative user experience seems to be more related to service design and platform management than to online dating as a concept.

In general, the combination of sentiment analysis and topic modeling proved useful in obtaining a more complete understanding of user reviews. Sentiment analysis identified the general emotional tendency of the reviews, while topic modeling explained which aspects of the apps generated those opinions. The results show that the main areas for improvement are transparency in account moderation, reduction of fake and scam profiles, clearer premium pricing, better customer support, more stable technical performance, and more reliable matching and messaging functionalities.

This study also has some limitations. First, the analysis is restricted to reviews from 2021, a period strongly influenced by the social effects of the pandemic and by the increased use of online interaction platforms. Second, sentiment analysis is based on a lexicon-based approach, which is simple and interpretable, but may not capture sarcasm, context, negation, or domain-specific meanings. Third, the aggregation from topics to macro-topics involved a manual interpretation step, which improves readability but also introduces a subjective component. Finally, app-store reviews can overrepresent users with particularly positive or negative experiences and therefore should not be interpreted as a perfect representation of the entire user base.

Future developments could extend the analysis to multiple years in order to study how user perceptions evolve over time. Moreover, more advanced sentiment models, such as supervised classifiers or transformer-based language models, could be used to capture contextual polarity with more precision. A further improvement would be to compare textual sentiment with numerical ratings, in order to verify whether the emotional tone of reviews is consistent with the star scores assigned by users.

In conclusion, the analysis shows that Tinder, Bumble, and Hinge face similar challenges in terms of user satisfaction, especially regarding monetization, trust, moderation, and platform reliability. However, the topic analysis highlights that each platform has its own specific weaknesses. For Hinge, dissatisfaction is mainly associated with fake users or bans, technical problems, and functionalities, while payments remain a frequent but more balanced topic. For dating apps, improving user experience does not only mean increasing the number of matches but also building a service perceived as transparent, fair, safe, technically reliable, and worth its cost.